

PROFESSIONAL SUMMARY

- AI Research Scientist at Meta with 6 years of post-PhD experience and 10+ years in AI research, spanning large-scale recommendation, large language models, and Trustworthy AI (calibration, alignment, fairness, interpretability)
- First author of Kunlun (KDD'26), an LLM-style architecture for ads late-stage ranking deployed in major Meta Ads models (MFU 17% to 37%, 2× scaling efficiency); revived the organic-to-ads (O2A) project (~30% offline top-1 gain) and now own it across 5+ teams and five launch paths; mentored 20+ students and junior researchers
- Published 50+ papers with 1,000+ Google Scholar citations at NeurIPS, ICML, ICLR, COLM, KDD, EMNLP, AAAI, UAI, TKDE, TNNLS, and more, including recent LLM calibration and alignment work (NeurIPS'26, COLM'26, ICML'25)
- Recognized with the *Distinguished Paper Award* (AMIA'24), *Best Paper Award* (ACM BCB'23, BCB'25), *PennAITech Innovation Fellow Award*, *Outstanding Student Paper Award* (CCFAI'19), and *Excellent Doctoral Dissertation Awards* (Jiangsu, JSAI, NJU)

EDUCATION

Nanjing University, Computer Science, *Ph.D.* (GPA: 4.46/5.00, 3/60), Nanjing, China Sep. 2014 - Jun. 2020

- Expertise in machine learning and artificial intelligence, LAMDA Group (advisor: [Prof. Zhi-Hua Zhou](#))

Nanjing University, Computer Science, *Bachelor of Science* (GPA: 4.33/5.00, 20/180), Nanjing, China Sep. 2010 - Jun. 2014

WORK EXPERIENCE

Meta, AI Research Scientist, Monetization & Meta Recommendation Systems (MRS), Menlo Park, CA Jul. 2025 - Present

- First-authored Kunlun (KDD'26), a ~30-author Meta Ads effort on an LLM-style unified architecture for late-stage ads ranking with predictable scaling laws; the architecture raises MFU from 17% to 37% on NVIDIA B200 GPUs, doubles scaling efficiency over state-of-the-art methods, and is deployed in major Meta Ads models
- Designed a request-only (RO) architecture for the production late-stage ranking model that computes request-level components once per request instead of once per candidate, delivering a 0.05% NE gain with no QPS regression
- Revived the organic-to-ads (O2A) representation project after a reorg left it with no offline gain and most contributors gone: introduced impressed-but-unclicked ads as hard negatives with an ordinal floor loss, improving offline top-1 accuracy by ~30%
- Own O2A end to end across 5+ partner teams (including data engineering and the Facebook recommendation product team): PyTorch contrastive modeling, Hive/Presto embedding pipelines, QRT pre-tests, and launch across five paths spanning user-to-user and ad-to-ad retrieval, ads boosting, feature injection, and sequence filtering
- Mentored a PhD research intern through a WSDM'27 submission on Directed Upstream Learning (adapting an upstream model for a frozen downstream recommender), a data engineer who built O2A training data under my guidance, and a new-hire teammate as officially assigned mentor; promoted about one year after joining
- Published LLM calibration research as corresponding author: bilevel fine-tuning that keeps aligned LLMs calibrated under domain shift where post-hoc temperature scaling fails (COLM'26), and a semantic-sampling calibration metric for open-ended QA with provably unbiased estimators, validated on five frontier LLMs (NeurIPS'26)
- Showed that length penalties fail to fix preference optimization because length is confounded with syntax ("syntactic shadowing"); counterfactual rewrites decompose the alignment gap into length, syntactic, and semantic parts, and across 25 models and ten methods only surface-matched pairs improve semantic preference (under review at ICLR'27)

University of Pennsylvania, Postdoctoral Fellow, Philadelphia, PA Jan. 2022 - Jul. 2025

- Mentored 20+ students through the full research cycle (problem formulation, coding, experiments, compute setup, paper writing, and rebuttals), resulting in co-authored papers at NeurIPS, ICML, ICLR, and KDD
- Showed that preference alignment degrades LLM calibration and developed calibration-aware fine-tuning (ICML'25); proposed attention-only fine-tuning that improves weight disentanglement for model merging via task arithmetic (ICLR'25)
- Built MentalChat16K, a counseling benchmark for LLMs (KDD'25 Oral); developed SEFD for paraphrase-robust detection of LLM-generated text (IEEE BigData'24) and DALK for LLM and knowledge-graph co-augmentation (EMNLP'24)
- Developed fairness-aware learning methods: Fair CCA (NeurIPS'23), fairness-aware graphical models (NeurIPS'24), and fairness-aware class-imbalanced learning (UAI'23); built interpretable deep clustering survival machines (MedIA'24)

Cornell University, Postdoctoral Associate, New York, NY May 2021 - Jan. 2022

- Developed a BERT-based clinical decision support system, improving literature retrieval accuracy by 4% and information extraction F1 by 5%; studied Shapley value stability for interpretability of medical AI

4Paradigm Corporation, Research Scientist, Beijing, China Aug. 2020 - Apr. 2021

- Pioneered "Environment Learning" for decision optimization, combining data-driven and knowledge-based approaches with reinforcement learning; virtual environment models cut validation loss by 79% for COVID-19 infection prediction

SELECTED PUBLICATIONS (1,000+ [Google Scholar](#) citations, * indicates co-first author, † indicates corresponding author)

[KDD'26] **Bojian Hou**, Xiaolong Liu, Xiaoyi Liu, Jiaqi Xu, Yasmine Badr, Mengyue Hang, et al. [Kunlun: Establishing Scaling Laws for Massive-Scale Recommendation Systems through Unified Architecture Design](#). In: Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (ADS Track), 2026.

- Unified architecture with predictable power-law scaling, combining GDPA, hierarchical seed pooling, sliding-window attention, computation skip, and event-level personalization; deployed in major Meta Ads models

[NeurIPS'26] Zhanliang Wang*, Jiancong Xiao*, Ruochen Jin, Shu Yang, **Bojian Hou**†, and Li Shen†. [A Semantic-Sampling Framework for Evaluating Calibration in Open-Ended Question Answering](#). In: Proceedings of the 40th Annual Conference on Neural Information Processing Systems (NeurIPS 2026), 2026, in press. arXiv:2605.08432.

- Sem-ECE: logit-free calibration metric for open-ended QA with unbiased estimators, validated on five frontier LLMs; the gap between same-sample and held-out estimators also diagnoses question difficulty

[COLM'26] Ruochen Jin*, Zhanliang Wang*, Zongyu Dai, Jiancong Xiao†, and **Bojian Hou**†. [Beyond Post-Hoc Temperature Scaling: Bilevel Optimization for LLM Calibration](#). In: Proceedings of the Conference on Language Modeling (COLM), 2026.

- CALM: first-order bilevel fine-tuning that keeps aligned LLMs calibrated under domain shift

[ICML'25] Jiancong Xiao*, **Bojian Hou***, Zhanliang Wang*, Ruochen Jin, Qi Long, Weijie J. Su, and Li Shen. [Restoring Calibration for Aligned Large Language Models: A Calibration-Aware Fine-Tuning Approach](#). In: Proceedings of the 42nd International Conference on Machine Learning (ICML), Vancouver, Canada, 2025.

- Showed preference alignment degrades LLM calibration; calibration-regularized fine-tuning restores it without hurting accuracy

[ICLR'25] Ruochen Jin*, **Bojian Hou***, Jiancong Xiao*, Weijie Su, and Li Shen. [Fine-Tuning Attention Modules Only: Enhancing Weight Disentanglement in Task Arithmetic](#). In: Proceedings of the 13th International Conference on Learning Representations, 2025.

- Fine-tuning only attention modules improves weight disentanglement and model merging via task arithmetic

[KDD'25] Jia Xu*, Tianyi Wei*, **Bojian Hou***, Patryk Orzechowski, Shu Yang, Ruochen Jin, Rachael Paulbeck, Joost Wagenaar, George Demiris, and Li Shen. [MentalChat16K: A Benchmark Dataset for Conversational Mental Health Assistance](#). In: Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2025. (Oral)

- Benchmark of synthetic and anonymized real counseling dialogues for fine-tuning and evaluating LLM mental-health assistants

[NeurIPS'25] Parvin Nazari, **Bojian Hou**, Davoud Ataee Tarzanagh, Li Shen, and George Michailidis. [Stochastic Regret Guarantees for Online Zeroth- and First-Order Bilevel Optimization](#). In: Advances in Neural Information Processing Systems 38, 2025.

- Regret guarantees for stochastic online bilevel optimization with zeroth- and first-order methods

[NeurIPS'24] Zhuoping Zhou*, Davoud A. Tarzanagh*, **Bojian Hou***, Qi Long, and Li Shen. [Fairness-Aware Estimation of Graphical Models](#). In: Advances in Neural Information Processing Systems 37, 2024.

- Fair graphical model estimation that reduces bias across protected groups while preserving model quality

[NeurIPS'23] Zhuoping Zhou*, Davoud A. Tarzanagh*, **Bojian Hou***, Boning Tong, Jia Xu, Yanbo Feng, Qi Long, and Li Shen. [Fair Canonical Correlation Analysis](#). In: Advances in Neural Information Processing Systems 36, 2023.

- Learns fair global projections by minimizing correlation disparity across protected groups while maintaining accuracy

[IEEE BigData'24] Weiqing He*, **Bojian Hou***, Tianqi Shang*, Davoud A. Tarzanagh, Qi Long, and Li Shen. [SEFD: Semantic-Enhanced Framework for Detecting LLM-Generated Text](#). In: 2024 IEEE International Conference on Big Data, 2024.

- Retrieval-based semantic framework that makes LLM-generated text detection robust to paraphrasing attacks

[UAI'23] Davoud A. Tarzanagh*, **Bojian Hou***, Boning Tong, Qi Long, and Li Shen. [Fairness-Aware Class Imbalanced Learning on Multiple Subgroups](#). In: Proceedings of the 39th Conference on Uncertainty in Artificial Intelligence, 2023, 2123-2133.

- Tri-level optimization for overparameterized models that improves generalization on imbalanced subgroups

[MedIA'24] **Bojian Hou***, Zixuan Wen*, Jingxuan Bao, Richard Zhang, Boning Tong, Shu Yang, Junhao Wen, Yuhan Cui, Jason H. Moore, Andrew J. Saykin, Heng Huang, Paul M. Thompson, Marylyn D. Ritchie, Christos Davatzikos, and Li Shen. [Interpretable Deep Clustering Survival Machines for Alzheimer's Disease Subtype Discovery](#). Medical Image Analysis, 2024, 97:103231.

- Interpretable model combining discriminative and generative components to discover subtypes and predict risk

Earlier first-author work on feature evolvable streams and interpretable RNNs: NeurIPS'17, IJCAI'17, TKDE'19, TNNLS'20/'21, AAAI'21.

TECHNICAL SKILLS

Methods: LLM fine-tuning, preference optimization and calibration, LLM-as-judge evaluation, attention architectures and scaling for ranking, contrastive representation learning with hard negatives, fairness, interpretability, bilevel optimization

Programming Languages: Python, Java, C/C++, SQL, MATLAB, Bash, LaTeX, HTML, CSS

Machine Learning Platform/Tools: PyTorch, FBLeaer Flow, Hugging Face, TensorFlow, Scikit-Learn, Pandas, Hive, Presto

ACADEMIC ACTIVITIES

(Senior) Program Committee of NeurIPS (2018-), ICML, ICLR, AAAI, AISTATS (2019-), KDD (2020-), and IJCAI (2021-)

Reviewer of Nature Methods, TPAMI, TNNLS, TII, Medical Image Analysis, Machine Learning, TKDD, and Scientific Reports